ECON 381: Practice Midterm 2 — Answers

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Fall 2026

Practice midterm answer key.

1. Multiple Choice Answers

Question	1	2	3	4	5	6	7	8	9	10
Answer	b	c	a	d	a	b	c	a	d	b

1. **(b)** A1 restricts the conditional error mean to zero at every education level. It does not require zero individual errors or constant error variance.

2. **(c)** More observations can reduce sampling uncertainty but do not eliminate systematic omitted determinants related to education. A1 remains a substantive condition.

3. **(a)** Shared county shocks threaten independence in A2. These facts alone do not establish whether A1, A3, or homoskedasticity holds.

4. **(d)** Under homoskedasticity, $\text{Var}(\hat{\beta}_1) \approx \text{Var}(u)/[n\text{Var}(X)]$. Replacing n by $4n$ divides the variance by 4 and the sampling SD by 2.

5. **(a)** The slope-variance ratio is $4/2 = 2$; the sampling-SD ratio is $\sqrt{2}$. Ratios refer to the large-sample population approximation, not a guaranteed ratio of two realized sample SEs.

6. **(b)** A Type-I error rejects a true null hypothesis.

7. **(c)** With a common SE and reference distribution, $|t| > 1.96$, $p < 0.05$, and exclusion of zero from the 95% CI are equivalent rules for this two-sided 5% test.

8. **(a)** $0.072 < 0.10$, but $0.072 > 0.05$. Both signs count in a two-sided test. The normal-approximation value for $|t| = 1.80$ is about 0.0719, consistent with the supplied p.

9. **(d)** $\beta_0 = E[Y \mid D = 0] = 660$ and $\beta_1 = E[Y \mid D = 1] - E[Y \mid D = 0] = 652 - 660 = -8$. This difference does not by itself establish a causal class-size effect.

10. **(b)** Changing conditional error variance is heteroskedasticity. A1 concerns the conditional mean, so it can hold at the same time.

2. True or False Answers

(a) **True.** Under the stated conditions, robust SEs allow either constant or changing conditional error variance. The `robust` option changes the uncertainty calculation, not the OLS coefficients or residuals.

(b) **False.** The 1% test uses a higher cutoff, 2.58 rather than 1.96, and requires stronger evidence. For example, $\hat{\beta}_1 = 0.44$ with $SE = 0.20$ gives $|t| = 2.20$: reject at 5%, but do not reject at 1%. Equivalently, its 95% CI $[0.05, 0.83]$ excludes zero, but its 99% CI $[-0.08, 0.96]$ includes zero.

(c) **False.** Residuals are calculated from the estimated sample line. Their zero sample mean follow from the OLS construction with an intercept; they do not reveal the *unobserved errors* from a proposed population or causal line. A1 concerns $E[u_i \mid X_i]$, which cannot be established directly from these residual identities.

(d) **True.** The SE uses one sample to estimate the sampling SD of the slope across repeated samples of the same size. That population SD is fixed for given n and population; the sample SE generally varies as the sample changes.

3. Analytical Answers

These calculations use the displayed two-decimal coefficients and standard errors and the provided critical values. Every calculated answer is reported to two decimal places.

(a) **Equation, SEs, and sample size.**

$$\widehat{\text{TestScore}}_i = 625.38 + 1.88\,\text{Income}_i$$
$$(1.87) \quad (0.11), \qquad n = 420.$$

Parentheses denote robust standard errors of the corresponding coefficients.

(b) **Slope and predictions.** A \$1,000 higher average district income is associated with a predicted average score higher by about 1.88 points. This is an association unless causal assumptions are justified.

$$\widehat{\text{TestScore}}(20) = 625.38 + 1.88(20) = \boxed{662.98}.$$

An income increase of \$5,000 corresponds to $\Delta\text{Income} = 5$:

$$\Delta\widehat{\text{TestScore}} = 1.88(5) = \boxed{9.40 \text{ points}}.$$

(c) **Intercept.** The fitted line predicts 625.38 test-score points at average district income of zero. Income zero lies outside the observed range (approximately \$5,340–\$55,330), so this is an extrapolation with no practical meaning. There are no school districts observed with zero income in the dataset.

(d) **SEs and t.** The intercept SE, 1.87, estimates the sampling SD of the intercept across repeated samples of 420 districts (in test-score points). The income SE, 0.11, estimates the sampling SD of the income slope (in score points per \$1,000 of income). They do not measure the dispersion of individual test scores or the regression errors.

$$t = \frac{1.88 - 0}{0.11} \approx \boxed{17.09}.$$

The estimated slope is about 17.09 estimated sampling SDs above zero.

Analytical Answers (continued)

(e) **Hypotheses, three rules, and interpretation.** $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$.

- **CI rule:** The reported 95% CI, $[1.66, 2.10]$, excludes zero, so reject at 5%. It is an interval for the population income slope, in score points per \$1,000, not for individual scores. Under valid assumptions, the procedure covers the fixed population slope in approximately 95% of repeated samples.
- t **rule:** $|t| = 17.09 > 1.96$, so reject using the handout's normal approximation.
- p **rule:** The displayed $p = 0.00$ is rounded, not exactly zero. It represents the two-sided probability, under a zero-slope null, of a statistic at least as extreme as the observed statistic. It is below 0.05, so reject. It is not the probability that H_0 is true.

Using the rounded inputs, $1.88 \pm 1.96(0.11)$ gives $[1.66, 2.10]$ after final rounding. The table's $t = 16.53$ was computed by Stata before display rounding; part (d) correctly recomputes $t = 17.09$ from $1.88/0.11$.

(f) **Confidence and significance levels.**

$$\begin{aligned} CI_{90\%} &= 1.88 \pm 1.65(0.11) \\ &\approx \boxed{[1.70, 2.06]}, \\ CI_{99\%} &= 1.88 \pm 2.58(0.11) \\ &\approx \boxed{[1.60, 2.16]}. \end{aligned}$$

Both intervals center on 1.88, and the 99% interval is wider. They exclude zero, so the slope is statistically different from zero at 10%, 5%, and 1%. Increasing confidence widens the interval; decreasing significance level raises the two-sided rejection cutoff (1.65, 1.96, 2.58) and requires stronger evidence. Here $|t|$ exceeds all three.

Analytical Answers (continued: assumptions)

(g) **Assumptions and limits.**

(i) **A1:** $E[u_i \mid \text{Income}_i = x] = 0$ at every x. For a causal income effect, omitted determinants must not vary with income in their contribution to scores. For example, parental educational attainment may influence scores and be systematically higher in richer districts. This makes A1 doubtful for the proposed causal interpretation. Zero sample residual mean is the consequence of using OLS and do not establish A1 for the unobserved population errors.

(ii) **A2:** Income–score pairs are independent and identically distributed. Sampling independently from a common target population supports it; shared county policies or economic shocks can threaten independence. The table does not describe sampling or establish i.i.d. observations. The 420 observed districts alone may not prove random sampling.

(iii) **A3:** Unusually large outliers are unlikely. Examine scatter plots, income distributions, and extreme observations. For example, unusually rich districts may warrant attention. A coefficient table cannot establish this population condition.

(iv) **Robust SEs:** Requesting them does not establish homoskedasticity. They allow either constant or changing conditional error variance, given the other assumptions and a reliable large-sample approximation. They do not repair a failure of A1 or dependence among districts. Statistical significance does not establish causality.

The table alone does not establish that any assumption definitively holds or fails.

4. Bonus Answers

(a) **Variance and SD ratios.** Under the stated conditions,

$$\mathrm{Var}(\hat{\beta}_1) \approx \frac{\mathrm{Var}(u)}{n\mathrm{Var}(X)}.$$

Write the Study A quantities as n_A, $\mathrm{Var}(u)_A$, and $\mathrm{Var}(X)_A$. Then

$$\frac{\mathrm{Var}(\hat{\beta}_{1,B})}{\mathrm{Var}(\hat{\beta}_{1,A})} \approx \frac{2\mathrm{Var}(u)_A}{(2n_A)(4\mathrm{Var}(X)_A)} \frac{n_A\mathrm{Var}(X)_A}{\mathrm{Var}(u)_A} = \frac{1}{4} = \boxed{0.25}.$$

The sampling-SD ratio is $\sqrt{1/4} = \boxed{0.50}$. Typical reported SEs approximately reflect this ratio in sufficiently large samples; individual realized SEs need not have exactly this ratio.

(b) **Causal interpretation. No.** Part (a)'s calculation assumed A1. Once A1 fails, the variance formula cannot simply be carried over as a valid variance around the causal effect. Even if a reported SE is small, sampling precision does not establish that the slope estimator targets the causal education effect. Ability systematically related to education remains a concern as n grows.